

Empirical Risk and Rademacher Complexity

Lecturer: Yun Wei

Scribe: Yuhang Zhao

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1 Empirical Risk Minimization

Suppose the observations have distribution P_{θ^*} , where $\theta^* \in \Omega$ is unknown. A loss function $L_{\theta}(X)$ measures the fit between a candidate parameter $\theta \in \Omega$ and an observation $X \in \mathcal{X}$. The population risk and empirical risk are, respectively,

$$R(\theta, \theta^*) := \mathbb{E}_{\theta^*}[L_{\theta}(X)], \quad (1)$$

$$\hat{R}_n(\theta, \theta^*) := \frac{1}{n} \sum_{i=1}^n L_{\theta}(X_i). \quad (2)$$

Given a nonempty candidate set $\Omega_0 \subseteq \Omega$, an empirical risk minimizer is any

$$\hat{\theta}_n \in \arg \min_{\theta \in \Omega_0} \hat{R}_n(\theta, \theta^*). \quad (3)$$

Its excess risk is

$$\mathbf{E}(\hat{\theta}_n, \theta^*) := R(\hat{\theta}_n, \theta^*) - \inf_{\theta \in \Omega_0} R(\theta, \theta^*). \quad (4)$$

Remark 1.1 (Correction to the binary classification example). In Example 4.9 of [1], the assumption $P(Y = +1) = P(Y = -1) = 1/2$ is unnecessary for the stated Bayes rule and should be removed.

For simplicity, assume the infimum in (4) is attained at a fixed parameter $\theta_0 \in \Omega_0$, so that

$$R(\theta_0, \theta^*) = \inf_{\theta \in \Omega_0} R(\theta, \theta^*).$$

Add and subtract the two empirical risks to obtain

$$\begin{aligned} \mathbf{E}(\hat{\theta}_n, \theta^*) &= R(\hat{\theta}_n, \theta^*) - R(\theta_0, \theta^*) \\ &= \underbrace{R(\hat{\theta}_n, \theta^*) - \hat{R}_n(\hat{\theta}_n, \theta^*)}_{T_1} \\ &\quad + \underbrace{\hat{R}_n(\hat{\theta}_n, \theta^*) - \hat{R}_n(\theta_0, \theta^*)}_{T_2 \leq 0} \\ &\quad + \underbrace{\hat{R}_n(\theta_0, \theta^*) - R(\theta_0, \theta^*)}_{T_3}. \end{aligned}$$

The term T_2 is nonpositive by the minimizing property of $\hat{\theta}_n$. The first term is

$$T_1 = \mathbb{E}_{\theta^*}[L_{\hat{\theta}_n}(X)] - \frac{1}{n} \sum_{i=1}^n L_{\hat{\theta}_n}(X_i).$$

To bound this term uniformly over the candidate parameters, define the loss class

$$\mathcal{L}(\Omega_0) := \{x \mapsto L_\theta(x) \mid \theta \in \Omega_0\}.$$

For any distribution P and a P -integrable function f , write

$$Pf := \mathbb{E}[f(X)], \quad P_n f := \frac{1}{n} \sum_{i=1}^n f(X_i),$$

where $X_1, \dots, X_n$ are i.i.d. from P . For a nonempty function class $\mathcal{F}$, the uniform deviation is

$$\|P_n - P\|_{\mathcal{F}} := \sup_{f \in \mathcal{F}} |P_n f - Pf| = \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n f(X_i) - \mathbb{E}[f(X)] \right|. \quad (5)$$

The displayed suprema and expectations are assumed to be well defined. With $P = P_{\theta^*}$, the two remaining terms satisfy

$$\begin{aligned} T_1 &= P_{\theta^*} L_{\hat{\theta}_n} - P_n L_{\hat{\theta}_n} \\ &\leq \sup_{\theta \in \Omega_0} |P_{\theta^*} L_\theta - P_n L_\theta| = \|P_n - P_{\theta^*}\|_{\mathcal{L}(\Omega_0)}, \\ T_3 &= P_n L_{\theta_0} - P_{\theta^*} L_{\theta_0} \leq \|P_n - P_{\theta^*}\|_{\mathcal{L}(\Omega_0)}. \end{aligned}$$

Combining these inequalities gives the following result.

Proposition 1.2 (Excess risk bound). *An empirical risk minimizer satisfies*

$$0 \leq \mathbf{E}(\hat{\theta}_n, \theta^*) \leq 2 \|P_n - P_{\theta^*}\|_{\mathcal{L}(\Omega_0)}. \quad (6)$$

This motivates the study of $\|P_n - P\|_{\mathcal{F}}$. Recall:

Definition 1.3 (Glivenko–Cantelli class). A function class $\mathcal{F}$ is *Glivenko–Cantelli* for P if

$$\|P_n - P\|_{\mathcal{F}} \longrightarrow 0 \quad \text{in probability as } n \rightarrow \infty.$$

It is *strong Glivenko–Cantelli* for P if this convergence holds almost surely.

2 A Uniform Law via Rademacher Complexity

In the remainder of the lecture, $X_1, \dots, X_n$ are i.i.d. from P , and $\mathcal{F}$ is a nonempty class of P -integrable functions.

Let $\varepsilon_1, \dots, \varepsilon_n$ be independent Rademacher variables, with

$$P(\varepsilon_i = +1) = P(\varepsilon_i = -1) = \frac{1}{2}.$$

They are independent of all observations. Recall that for a nonempty set $A \subseteq \mathbb{R}^n$, its Rademacher complexity is

$$\mathcal{R}(A) := \mathbb{E}_\varepsilon \left[\sup_{a \in A} |\langle a, \varepsilon \rangle| \right] = \mathbb{E}_\varepsilon \left[\sup_{a \in A} \left| \sum_{i=1}^n a_i \varepsilon_i \right| \right]. \quad (7)$$

For a fixed collection $x_1^n = (x_1, \dots, x_n)$, define

$$\mathcal{F}(x_1^n) := \{(f(x_1), \dots, f(x_n)) \mid f \in \mathcal{F}\} \subseteq \mathbb{R}^n.$$

These are all the vectors that can be obtained by evaluating functions in $\mathcal{F}$ on the same collection of points.

Definition 2.1 (Empirical and population Rademacher complexity). The *empirical Rademacher complexity* on x_1^n is

$$\mathcal{R}(\mathcal{F}(x_1^n)/n) := \mathbb{E}_\varepsilon \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(x_i) \right| \right]. \quad (8)$$

For the random sample $X_1^n = (X_1, \dots, X_n)$, the *Rademacher complexity of the function class* is

$$\begin{aligned} \mathcal{R}_n(\mathcal{F}) &:= \mathbb{E}_X [\mathcal{R}(\mathcal{F}(X_1^n)/n)] \\ &= \mathbb{E}_{X, \varepsilon} \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right| \right]. \end{aligned} \quad (9)$$

We first recall the symmetrization inequality for random vectors.

Theorem 2.2 (Symmetrization for mean-zero random vectors). *Let $X_1, \dots, X_n$ be independent, mean-zero random vectors in a normed space. Let $\varepsilon_1, \dots, \varepsilon_n$ be independent Rademacher variables, independent of the vectors. Then*

$$\frac{1}{2} \mathbb{E}_{X, \varepsilon} \left\| \sum_{i=1}^n \varepsilon_i X_i \right\| \leq \mathbb{E}_X \left\| \sum_{i=1}^n X_i \right\| \leq 2 \mathbb{E}_{X, \varepsilon} \left\| \sum_{i=1}^n \varepsilon_i X_i \right\|. \quad (10)$$

Homework 1: Let $X_1, \dots, X_n$ be independent random vectors in a normed space, and let $\varepsilon_1, \dots, \varepsilon_n$ be independent Rademacher variables independent of these vectors. Then

$$\mathbb{E}_X \left\| \sum_{i=1}^n (X_i - \mathbb{E}X_i) \right\| \leq 2 \mathbb{E}_{X, \varepsilon} \left\| \sum_{i=1}^n \varepsilon_i X_i \right\|.$$

Combining the above two immediately yields:

Theorem 2.3 (Symmetrization for general independent random vectors). *Let $X_1, \dots, X_n$ be independent, random vectors in a normed space. Let $\varepsilon_1, \dots, \varepsilon_n$ be independent Rademacher variables, independent of these vectors. Then*

$$\begin{aligned} \frac{1}{2} \mathbb{E}_{X, \varepsilon} \left\| \sum_{i=1}^n \varepsilon_i (X_i - \mathbb{E}X_i) \right\| &\leq \mathbb{E}_X \left\| \sum_{i=1}^n (X_i - \mathbb{E}X_i) \right\| \\ &\leq 2 \mathbb{E}_{X, \varepsilon} \left\| \sum_{i=1}^n \varepsilon_i X_i \right\|. \end{aligned} \quad (11)$$

Lemma 2.4 (Bounds on the mean uniform deviation). *For a class $\mathcal{F}$ of P -integrable functions,*

$$(a) \quad \mathbb{E} [\|P_n - P\|_{\mathcal{F}}] \leq 2 \mathcal{R}_n(\mathcal{F}). \quad (12)$$

$$(b) \quad \mathbb{E} [\|P_n - P\|_{\mathcal{F}}] \geq \frac{1}{2} \mathcal{R}_n(\overline{\mathcal{F}}), \quad (13)$$

where

$$\overline{\mathcal{F}} := \{f - Pf \mid f \in \mathcal{F}\}$$

is the recentered function class.

Proof. (a)

$$\begin{aligned} \|P_n - P\|_{\mathcal{F}} &= \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n f(X_i) - \mathbb{E}[f(X)] \right| \\ &= \sup_{f \in \mathcal{F}} \left| \sum_{i=1}^n \frac{1}{n} (f(X_i) - \mathbb{E}[f(X_i)]) \right|. \end{aligned}$$

Apply (11) to the vectors $(f(X_i)/n)_{f \in \mathcal{F}}$ with the supremum norm over $\mathcal{F}$. The upper bound gives

$$\begin{aligned}\mathbb{E} [\|P_n - P\|_{\mathcal{F}}] &\leq 2\mathbb{E}_{X,\varepsilon} \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right| \right] \\ &= 2\mathcal{R}_n(\mathcal{F}).\end{aligned}$$

(b) The lower bound in (11) gives

$$\begin{aligned}\mathbb{E} [\|P_n - P\|_{\mathcal{F}}] &\geq \frac{1}{2}\mathbb{E}_{X,\varepsilon} \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i (f(X_i) - \mathbb{E}[f(X_i)]) \right| \right] \\ &= \frac{1}{2}\mathcal{R}_n(\overline{\mathcal{F}}).\end{aligned}$$

□

Remark 2.5 (Convergence in mean and in probability). If $\mathcal{R}_n(\mathcal{F}) = o(1)$, then (12) implies

$$\mathbb{E} [\|P_n - P\|_{\mathcal{F}}] \longrightarrow 0.$$

Consequently,

$$\|P_n - P\|_{\mathcal{F}} \longrightarrow 0 \quad \text{in } L^1 \text{ and in probability.}$$

Thus $\mathcal{F}$ is Glivenko–Cantelli for P .

Definition 2.6 (Uniformly bounded function class). A class $\mathcal{F}$ is *b-uniformly bounded* if

$$\|f\|_{\infty} := \sup_{x \in \mathcal{X}} |f(x)| \leq b \quad \text{for every } f \in \mathcal{F}.$$

Theorem 2.7 (Uniform law via Rademacher complexity). *For a b-uniformly bounded class $\mathcal{F}$ with $b > 0$, every $n \geq 1$ and $\delta \geq 0$ satisfy*

$$\|P_n - P\|_{\mathcal{F}} \leq 2\mathcal{R}_n(\mathcal{F}) + \delta \tag{14}$$

with probability at least

$$1 - \exp\left(-\frac{n\delta^2}{2b^2}\right).$$

Recall the following consequence of the first Borel–Cantelli lemma.

Lemma 2.8 (A criterion for almost sure convergence). *If random variables X_n and X satisfy*

$$\sum_{n=1}^{\infty} P(|X_n - X| > \epsilon) < \infty \quad \text{for every } \epsilon > 0,$$

then $X_n \rightarrow X$ almost surely.

Lemma 2.9. *If random variables X_n satisfy*

$$\sum_{n=1}^{\infty} P(X_n > \epsilon) < \infty \quad \text{for every } \epsilon > 0,$$

then $\limsup_n X_n \leq 0$ almost surely.

Corollary 2.10 (Strong Glivenko–Cantelli property). *If $\mathcal{F}$ is b-uniformly bounded and $\mathcal{R}_n(\mathcal{F}) = o(1)$, then*

$$\|P_n - P\|_{\mathcal{F}} \longrightarrow 0 \quad \text{almost surely.}$$

Proof. The case $b = 0$ is immediate. Suppose $b > 0$ and fix $\delta > 0$. Theorem 2.7 gives

$$\begin{aligned} \sum_{n=1}^{\infty} P(\|P_n - P\|_{\mathcal{F}} - 2\mathcal{R}_n(\mathcal{F}) > \delta) &\leq \sum_{n=1}^{\infty} \exp\left(-\frac{n\delta^2}{2b^2}\right) \\ &= \sum_{n=1}^{\infty} \left(e^{-\delta^2/(2b^2)}\right)^n < \infty. \end{aligned}$$

By Lemma 2.9,

$$\limsup_{n \rightarrow \infty} [\|P_n - P\|_{\mathcal{F}} - 2\mathcal{R}_n(\mathcal{F})] \leq 0 \quad \text{almost surely.}$$

Since $\mathcal{R}_n(\mathcal{F}) \rightarrow 0$ and $\|P_n - P\|_{\mathcal{F}} \geq 0$, it follows that $\|P_n - P\|_{\mathcal{F}} \rightarrow 0$ almost surely. $\square$

Proof of Theorem 2.7. For each $f \in \mathcal{F}$, write $\bar{f}(x) := f(x) - Pf$. Then

$$\begin{aligned} \|P_n - P\|_{\mathcal{F}} &= \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n (f(X_i) - Pf) \right| \\ &= \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \bar{f}(X_i) \right|. \end{aligned}$$

Define

$$G(x_1, \dots, x_n) := \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \bar{f}(x_i) \right|.$$

Thus $G(X_1, \dots, X_n) = \|P_n - P\|_{\mathcal{F}}$. Suppose $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$ differ only in coordinate k . Then

$$\begin{aligned} |G(x) - G(y)| &\leq \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \bar{f}(x_i) - \frac{1}{n} \sum_{i=1}^n \bar{f}(y_i) \right| \\ &= \frac{1}{n} \sup_{f \in \mathcal{F}} |\bar{f}(x_k) - \bar{f}(y_k)| \\ &= \frac{1}{n} \sup_{f \in \mathcal{F}} |f(x_k) - f(y_k)| \leq \frac{2b}{n}. \end{aligned}$$

Thus G has bounded difference constants $L_k = 2b/n$ for every k .

The one-sided bounded differences inequality yields

$$\begin{aligned} P(G(X_1^n) - \mathbb{E}G(X_1^n) > \delta) &\leq \exp\left(-\frac{2\delta^2}{\sum_{k=1}^n (2b/n)^2}\right) \\ &= \exp\left(-\frac{n\delta^2}{2b^2}\right). \end{aligned}$$

By (12), $\mathbb{E}G(X_1^n) \leq 2\mathcal{R}_n(\mathcal{F})$. Combining the mean bound with the concentration inequality proves (14). $\square$

References

- [1] M. J. Wainwright, *High-Dimensional Statistics: A Non-Asymptotic Viewpoint*, Cambridge University Press, 2019. Sections 4.1–4.2, pp. 102–108.